Career Launcher MBA

QA Arithmetic – 11

Circular Races; To and Fro

Question 1

A sprinter starts running on a circular path of radius r metres. Her average speed (in metres/minute) is πr during the first 30 seconds, $\pi r/2$ during next one minute, $\pi r/4$ during next 2 minutes, $\pi r/8$ during next 4 minutes and so on. What is the ratio of the time taken for the nth round to that for the (n – 1)th round?

(n > 1)

(1) 4 : 1

(2) 8 : 1

(3) 16 : 1

(4) 32 : 1

Question 2

A and B start from a point P on a circular track of length 280 m, simultaneously at speeds of 7 m/s and 4 m/s. Find

(1) The time after the start when they would meet for the first time and the third time if they run in the same direction.

(2) The time after the start when they would meet for the first time and the third time if they run in the opposite directions.

(3) Meet for the first time at P after the start, if they are both moving in the same direction.

(4) Number of points where they meet on the circular track when they are running in opposite directions.

Question 3

A, B, C start simultaneously from P on a circular track of length 280 m, at speeds of 7 m/s, 3 m/s, and 4 m/s respectively. When would they meet for the first time if

(1) A, B, C are moving in clockwise direction.

(2) A, B are moving in the clockwise direction, whereas C is moving in the anticlockwise direction.

Question 4

Two boys start simultaneously from the same point on a circular track and run in opposite directions at speeds of 2.25 km/hr and 3.5 km/hr respectively. If circumference of the track is 736 m, after how much time they meet for the first time?

(1) 7.5 min

(2) 7.68 min

(3) 8 min

(4) 6.86 min

Question 5

Amol and Avash start running simultaneously around a circular track from the same point in the same direction. Amol can complete 1 round in 16 seconds and Avash can do the same in 28 seconds. Find the time after which they would be together for the first time.

(1) 40 s

(2) $\frac{112}{3}$ s

(3) $27\frac{2}{3}$ s

(4) 37 s

Question 6

Two runners A and B are running on a circular track of length 14 km. If A and B started simultaneously from the same point but in opposite directions with speeds 15m/s and 20m/s respectively. What is the distance covered by A by the time they cross each other for the first time?

Question 7

A and B run a 5 km race on a round course of 400 m. If their speeds are in the ratio 5 : 4, then how many times will the winner of the race cross the other participant?

Question 8

A and B run along a circular track, of unknown radius, in the same direction at speeds of 3m/sec and 5m/sec respectively. Find at how many points will they meet.

(1) 2

(2) 3

(3) 1

(4) Cannot be determined

Question 9

Two persons start cycling simultaneously from a point P on a circular track. If they travel at a speeds of 3 km/hr and 10 km/hr respectively, then what is the ratio of number of distinct points that they would meet on that track then when they travel in the same direction first to that when they travel in the opposite direction?

(1) 7 : 13

(2) 3 : 10

(3) 3 : 7

(4) 30 : 13

Question 10

In a 4000 meter race around a circular stadium having a circumference of 1000 meters, the fastest runner and the slowest runner reach the same point at the end of the 5th minute, for the first time after the start of the race. All runners have the same starting point and each runner maintains a uniform speed throughout the race. If fastest runner runs at twice the speed of slowest runner, what is the time taken by fastest runner to finish the race?

For Questions 11 and 12

Answer the questions on the basis of the information given below.

Chiku and Charu are running along a circular path with a diameter of 1050 m at the speeds of 15 m/s and 25 m/s respectively. Chaman is running along the path AB (diameter of the circular path) at the speed of 35 m/s. Chaman turns back immediately after reaching the point B and does the same as he reaches the point A. All of them start from point A simultaneously.

11. If Charu and Chiku are running in the opposite direction, what is the shortest distance (in metres) along the circular path between point A and the point at which Charu and Chiku meet for the third time?

(1) 247.5

(2) 412.5

(3) 660

(4) 1320

12. If Charu and Chiku are running in the opposite direction, what is the shortest distance (in metres) between Charu and Chiku along the circular path when Chaman reaches the point B for the fifth time?

(1) 150

(2) 270

(3) 412.5

(4) 900

Question 13

Two ants A and B start from a point P on a circle at the same time, with A moving clock-wise and B moving anti-clockwise. They meet for the first time at 10:00 am when A has covered 60% of the track. If A returns to P at 10:12 am, then B returns to P at

(1) 10:18 am

(2) 10:27 am

(3) 10:25 am

(4) 10:45 am

Question 14

Two circular tracks T1 and T2 of radii 100 m and 20 m, respectively touch at a point A. Starting from A at the same time, Ram and Rahim are walking on track T1 and track T2 at speeds 15 km/hr and 5 km/hr respectively. The number of full rounds that Ram will make before he meets Rahim again for the first time is

(1) 4

(2) 2

(3) 5

(4) 3

Question 15

Anil, Sunil, and Ravi run along a circular path of length 3 km, starting from the same point at the same time, and going in the clockwise direction. If they run at speeds of 15 km/hr, 10 km/hr, and 8 km/hr, respectively, how much distance in km will Ravi have run when Anil and Sunil meet again for the first time at the starting point?

(1) 4.2

(2) 4.8

(3) 5.2

(4) 4.6

Question 16

A and B start from one end of a 50 m pool simultaneously, during a 1 km race. The speeds of A and B are 5 m/s and 3 m/s.

(1) After the start after how much time are A, B at any end of the pool again?

(2) How many times do they meet after the start, during the race?

Question 17

A and B start from the opposite ends of a 50 m pool simultaneously, during a 1 km race. The speeds of A and B are 5 m/s and 4 m/s.

(1) After the start after how much time are A, B at any end of the pool again?

(2) How many times do they meet after the start, during the race?

Question 18

A and B start from one end of a 50 m pool simultaneously, during a 1 km race. The speeds of A and B are 5 m/s and 3 m/s. When do they meet the 1st , 2nd and 7th time after the start?

Thanks!